



Assignments

2D Arrays – 2



Q1. Write a program to print the elements of both the diagonals in a square matrix.

Input 1:

1 2 3

4 5 6

7 8 9

Output 1:

1 3

5

7 9

Solution :

```
#include<iostream>
using namespace std;
int main(){
    int n ;
    cout << "Enter the number of rows : ";
    cin >> n;
    int arr[n][n];
    cout << "Enter the elements of matrix : "<<endl;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++)cin>>arr[i][j];
    }
    cout << "Elements of both the diagonals are as follows : 
    "<<endl;
    for(int i = 0 ; i < n ; i++){
        for(int j = 0 ; j < n ; j++){
            if((i + j == n - 1) or (i == j))cout << arr[i][j] << " ";
            else cout << " ";
        }
        cout<<endl;
    }
}
```

Q2. Write a program to rotate the matrix by 90 degrees anti-clockwise.

Input 1:

1 2 3

4 5 6

7 8 9

Output 1:

3 6 9

2 5 8

1 4 7

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n;
    cin>>n;
    int a[n][n];
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++)cin>>a[i][j];
    }
    // let's first calculate the transpose of the given matrix
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i <= j)swap(a[i][j] , a[j][i]);
        }
    }
    for(int j=0;j<n;j++){
        for(int i=0;i<n/2;i++){
            swap(a[i][j] , a[n-i-1][j]);
        }
    }
}
```

Q3. Write a program to print the matrix in wave form.

Input 1:

1 2 3

4 5 6

7 8 9

Output 1: 7 4 1 2 5 8 9 6 3

Solution :

```
#include<iostream>
using namespace std;
int main(){
    int n , m;
    cout << "Enter the number of rows : ";
    cin >> n;
    cout << "Enter the number of columns : ";
```

```
cin >> m;
int a[n][n];
cout << "Enter the matrix elements : "<<endl;
for(int i = 0 ; i < n ; i++){
for(int j = 0 ; j < m ; j++){
cin >> a[i][j];
}
}
cout<<"Elements in the wave form are: "<<endl;
for(int j = 0 ; j < m ; j++){
```

Q4. Given a positive integer n, generate a n x n matrix filled with elements from 1 to n² in spiral order.

Input 1: n = 3

Output 1: [[1,2,3],[8,9,4],[7,6,5]]

Input 2: n = 1

Output 2: [[1]]

Solution :

```
#include<iostream>
using namespace std;
int main(){
int n ;
cout << "Enter the number of rows : ";
cin >> n;
int arr[n][n];
int k = 1, i = 0;
while( k ≤ n * n ){
int j = i;
// four steps
while( j < n - i ) // 1. horizontal, left to right
arr[i][j++] = k++;
j = i + 1;
while( j < n - i ) // 2. vertical, top to bottom
arr[j++][n-i-1] = k++;
j = n - i - 2;
while( j > i ) // 3. horizontal, right to left
arr[n-i-1][j--] = k++;
j = n - i - 1;
while( j > i ) // 4. vertical, bottom to top
arr[j--][i] = k++;
```

Q5. Predict the output :

```
int main(){  
int a[][2] = {{1,2},{3,4}};  
int i, j;  
for (i = 0; i < 2; i++)  
for (j = 0; j < 2; j++)  
cout << a[i][j];  
return 0;  
}
```

Output : 1234



SKILLS



**THANK
YOU !**

